

Bill of Materials - VBF-T6LC-BOM-01

BOM

Component	Material	Mass (g/cell)	kg/kWh	Source note
Positive active material	LNMO LiNi0.5Mn1.5O4	17.13	0.859	parameter set: thickness x area x volume fraction x density
Positive conductive additive	carbon black (lit. 2 wt%)	0.36	0.018	literature default (not modeled in parameter set)
Positive binder	PVDF (lit. 2 wt%)	0.36	0.018	literature default (not modeled in parameter set)
Negative active material	graphite	14.04	0.704	parameter set
Negative conductive additive	carbon black (lit. 2 wt%)	0.29	0.015	literature default (not modeled)
Negative binder	SBR/CMC (lit. 2 wt%)	0.29	0.015	literature default (not modeled)
Separator	polyolefin	0.674	0.0338	parameter set: thickness x area x (1-porosity) x density
Electrolyte	EC/EMC + LiPF6	6.21	0.311	pore volume x 1.2 g/cm3 (literature density)
Positive current collector	Al 10 um	2.77	0.139	parameter set
Negative current collector	Cu 8 um	7.34	0.368	parameter set
Enclosure	pouch/housing	Not modeled	Not modeled	outside pure-simulation boundary
Tabs	Al/Ni/Cu	Not modeled	Not modeled	outside pure-simulation boundary
Total (stack, excl. electrolyte/additives)	None	41.96	2.105	calc-energy mass_kg caliber
Total (incl. electrolyte + binder/conductive)	None	49.46	2.481	annotated